# Explicit variety-evasive subspace families for bounded-degree projective and affine varieties

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doi:10.1007/s00037-024-00256-1; CC BY 4.0

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*Allowed standard prerequisites (editorial).*

- Projective algebraic varieties, Grassmannians/Plucker coordinates, Chow embeddings and the standard dimension formula for Chow varieties. Original definitions and citedTheorems3.5/3.7/4.1/4.4 retained, with author-provided explanatory argument. Classical results and projective hyperplane dimension fact cited rather than recursively reproved.

## Original ambient variety and evasion definitions (journal pp.3–6)

Let $\mathbb{F}$ be an algebraically closed field. An affine $n$-space $\mathbb{A}^n$, as a set, is simply defined to be the vector space $\mathbb{F}^n$. We also need the notion of a projective $n$-space, denoted by $\mathbb{P}^n$, which is (intuitively) the set of lines passing through the origin 0 of $\mathbb{A}^{n+1}$. Formally, it is defined to be the quotient set $(\mathbb{A}^{n+1} \setminus \{0\})/\sim$, where $\sim$ is the equivalence relation defined by scaling, i.e., $u \sim v$ if $u = cv$ for some nonzero scalar $c \in \mathbb{F}$. An (affine) subvariety $\mathcal{V} \subseteq \mathbb{A}^n$ is the set of common zeros of a set of $n$-variate polynomials over $\mathbb{F}$. Similarly, a (projective) subvariety $\mathcal{V} \subseteq \mathbb{P}^n$ is the set of common zeros of a set of homogeneous $(n+1)$-variate polynomials over $\mathbb{F}$, where we represent each element of $\mathbb{P}^n$ as an $(n+1)$-tuple in $\mathbb{A}^{n+1}$. In this paper, a variety refers to a subvariety of a projective or affine space, and is said to be irreducible if it cannot be written as a union of finitely many proper subvarieties. (Varieties in this paper are not necessarily irreducible. They are often called algebraic sets in literature.) The dimension of

a variety $\mathcal{V}$, denoted by $\dim(\mathcal{V})$, is intuitively the "degree of freedom" of picking a point in the variety. See Section 2.3 for its formal definition. In particular, for a linear subspace $\mathcal{V} \subseteq \mathbb{A}^n$, the dimension of $\mathcal{V}$ as a variety is simply its linear-algebraic dimension.

**Definition 1.1 (Evading).** *Let $\mathcal{V}_1$ and $\mathcal{V}_2$ be irreducible subvarieties of $\mathbb{P}^n$ or $\mathbb{A}^n$. We say $\mathcal{V}_1$ evades $\mathcal{V}_2$ if*

$$\dim(\mathcal{V}_1 \cap \mathcal{V}_2) \leq \dim(\mathcal{V}_1) + \dim(\mathcal{V}_2) - n,$$

*where the dimension of an empty set is assumed to be $-\infty$. In particular, if $\dim(\mathcal{V}_1) + \dim(\mathcal{V}_2) < n$, then $\mathcal{V}_1$ evades $\mathcal{V}_2$ iff $\mathcal{V}_1 \cap \mathcal{V}_2 = \emptyset$. More generally, suppose $\mathcal{V}_1$ is irreducible but $\mathcal{V}_2$ is possibly reducible. We say $\mathcal{V}_1$ evades $\mathcal{V}_2$ if it evades every irreducible component of $\mathcal{V}_2$.*

**Definition 1.2 (Subspace family).** *For $0 \leq k \leq n$, a finite collection (i.e., multiset) of $k$-subspaces of $\mathbb{P}^n$ is called a (projective) $k$-subspace family on $\mathbb{P}^n$. Similarly, a finite collection of affine $k$-subspaces of $\mathbb{A}^n$ is called an affine $k$-subspace family on $\mathbb{A}^n$.*

**Definition 1.3 (Variety evasive subspace family).** *Let $\mathcal{F}$ be a family of subvarieties of $\mathbb{P}^n$ (resp. $\mathbb{A}^n$). Let $\mathcal{H}$ be a $k$-subspace family on $\mathbb{P}^n$ (resp. affine $k$-subspace family on $\mathbb{A}^n$) where $0 \leq k \leq n$. Then:*

- *We say $\mathcal{H}$ is $\mathcal{F}$-evasive if for every $\mathcal{V} \in \mathcal{F}$, there exists $W \in \mathcal{H}$ that evades $\mathcal{V}$.*
- *We say $\mathcal{H}$ is $(\mathcal{F}, \epsilon)$-evasive if for every $\mathcal{V} \in \mathcal{F}$, a random element $W \in \mathcal{H}$ evades $\mathcal{V}$ with probability at least $1 - \epsilon$.*

**Definition 1.4 (degree).** *The degree of an irreducible variety $\mathcal{V}$ in $\mathbb{P}^n$ (resp. $\mathbb{A}^n$) is the number of intersections of $\mathcal{V}$ with a general projective (resp. affine) subspace of codimension $\dim(\mathcal{V})$. Following Heintz (1983); Heintz & Schnorr (1980), we define the degree of a (possibly reducible) variety to be the sum of the degrees of its irreducible components.*

# Original Plücker and Chow-form background (journal pp.27–30)

**Grassmannians.** Let $n \in \mathbb{N}$ and $k \in \{0, 1, \ldots, n-1\}$. The Grassmannian $\mathrm{G}(k+1, n+1)$ is the set of all $(k+1)$-dimensional linear subspaces of $\mathbb{A}^{n+1}$. By taking the quotient modulo scalars, it may also be identified with the set of all $k$-subspaces of $\mathbb{P}^n$, which we denote by $\mathrm{G}(k, n)$.

**The Plücker embedding and Plücker coordinates.** Consider a linear subspace $W \in \mathrm{G}(k+1, n+1)$. The simplest way of representing $W$ is using a $(k+1) \times (n+1)$ matrix $A$ over $\mathbb{F}$ such that $W$ equals the row space of $A$. We call such a matrix $A$ a generating matrix of $W$. For convenience, we also say $A$ is a generating matrix of $\mathbb{P}W \in \mathrm{G}(k, n)$.

**Definition 3.4 (Plücker embedding).** *Define* $\phi : \mathrm{G}(k+1, n+1) \to \mathbb{P}^{\binom{n+1}{k+1}-1}$ *by*

$$\phi(W) = \left(\det A_{[k+1],S}\right)_{S \in \binom{[n+1]}{k+1}}$$

*where* $A$ *is a generating matrix of* $W$*.*

The Plücker embedding embeds the Grassmannian $\mathrm{G}(k+1, n+1)$ in $\mathbb{P}^{\binom{n+1}{k+1}-1}$ as an irreducible projective subvariety, as stated by the following theorem. See, e.g., Fulton (1997); Harris (1992) for proofs.

**Theorem 3.5.** *The Plücker embedding* $\phi$ *is a well-defined injective map whose image is an irreducible projective subvariety of* $\mathbb{P}^{\binom{n+1}{k+1}-1}$*.*

The homogeneous coordinates $\left(\det A_{[k+1],S}\right)_{S \in \binom{[n+1]}{k+1}}$ of $\phi(W)$ are called the (primal) Plücker coordinates of $W$. Denote by $R := \mathbb{F}[X_S : S \in \binom{[n+1]}{k+1}]$ the homogeneous coordinate ring of $\mathbb{P}^{\binom{n+1}{k+1}-1}$. The irreducible projective subvariety $\phi(\mathrm{G}(k+1, n+1))$ is defined by a homogeneous prime ideal of $R$, which we denoted by $I$. Then $R/I$ is the homogeneous coordinate ring of $\phi(\mathrm{G}(k+1, n+1))$. The ideal $I$ contains precisely the polynomial relations that the Plücker coordinates need to satisfy. It is also known that $I$ is generated by certain quadratic forms, known as the Plücker relations. See Fulton (1997); Harris (1992) for details.

**Dual Plücker coordinates.** Alternatively, we could represent a linear subspace $W \in \mathrm{G}(k+1, n+1)$ by an $(n-k) \times (n+1)$ matrix $B$ over $\mathbb{F}$ whose rows specify the linear equations defining $W$. We call such a matrix $B$ a parity check matrix of $W$. For convenience, we also say $B$ is a parity check matrix of $\mathbb{P}W \in \mathrm{G}(k, n)$. The entries of $B$ are called the dual Stiefel coordinates of $W$. This gives another embedding $\phi^\vee : \mathrm{G}(k+1, n+1) \to \mathbb{P}^{\binom{n+1}{n-k}-1} = \mathbb{P}^{\binom{n+1}{k+1}-1}$, defined by

$$\phi^\vee(W) = \left(\det B_{[n-k],S}\right)_{S \in \binom{[n+1]}{n-k}}.$$

The homogeneous coordinates $\left(\det B_{[n-k],S}\right)_{S \in \binom{[n+1]}{n-k}}$ of $\phi^\vee(W)$ are called the dual Plücker coordinates of $W$. Recall that we denote by $\mathrm{G}(k, n)$ the set of all $k$-subspaces of $\mathbb{P}^n$. By identifying $\mathrm{G}(k+1, n+1)$ with $\mathrm{G}(k, n)$ via $W \mapsto \mathbb{P}W$, we regard $\phi$ and $\phi^\vee$ as maps from $\mathrm{G}(k, n)$ to $\mathbb{P}^{\binom{n+1}{k+1}-1}$.

**Definition 3.6 (Gelfand et al. 1994).** *For an irreducible subvariety* $\mathcal{V} \subseteq \mathbb{P}^n$ *of dimension* $n-k-1$*, define the associated hypersurfaces* $Z_\mathcal{V}$ *of* $\mathcal{V}$ *as the set of* $k$*-subspaces intersecting* $\mathcal{V}$*, i.e.,*

$$Z_\mathcal{V} := \{W \in \mathrm{G}(k, n) : \mathcal{V} \cap W \neq \emptyset\}.$$

The term "associated hypersurface" is justified by the following theorem.

**Theorem 3.7.** *Let* $\mathcal{V} \subseteq \mathbb{P}^n$ *be an irreducible projective subvariety of dimension* $n-k-1$ *and degree* $d \in \mathbb{N}^+$*. Then there exists a nonzero homogeneous polynomial* $P_\mathcal{V} \in R = \mathbb{F}[X_S : S \in \binom{[n+1]}{k+1}]$ *of degree* $d$ *such that* $\phi(Z_\mathcal{V})$ *is defined by* $P_\mathcal{V}$ *as a subvariety of* $\phi(\mathrm{G}(k, n))$*. That is,*

$$\phi(Z_\mathcal{V}) = \phi(\mathrm{G}(k, n)) \cap V(P_\mathcal{V}).$$

*Moreover, $R_{\mathcal{V}} := P_{\mathcal{V}} + I \in (R/I)_d$ is uniquely determined by $\mathcal{V}$ up to scalars.*

Theorem 3.7 is explicitly stated as Dalbec & Sturmfels (1995, Theorem 1.1 and Corollary 2.1). A proof can be found in Gelfand et al. (1994, Section 3.2). We briefly explain how to find a polynomial $P_{\mathcal{V}}$ satisfying Theorem 3.7: Firstly, it can be shown using the trick of dimension counting via incidence varieties that $\phi(Z_{\mathcal{V}})$ is an irreducible projective subvariety of the Grassmannian $\phi(\mathrm{G}(k,n))$ of codimension one (Gelfand et al. 1994, Section 3.2, Proposition 2.2). Secondly, the homogeneous coordinate ring $R/I$ of the Grassmannian is known to be a unique factorization domain (Fulton 1997, Chapter 9). These two facts imply that the homogeneous ideal of $R/I$ defining $\phi(Z_{\mathcal{V}})$ is a principal ideal. Choose $R_{\mathcal{V}}$ to be a generator of this principal ideal, which is unique up to scalars. Then lift $R_{\mathcal{V}} \in R/I$ to $P_{\mathcal{V}} \in R$.

**Definition 3.8 (Chow form).** *Let $\mathcal{V} \subseteq \mathbb{P}^n$ be an irreducible subvariety of dimension $n-k-1$ and degree $d \in \mathbb{N}^+$. Define the Chow form of $\mathcal{V}$ in Plücker coordinates, or simply the Chow form of $\mathcal{V}$, to be $R_{\mathcal{V}} \in (R/I)_d$ as in Theorem 3.7. More generally, for an equidimensional subvariety $\mathcal{V} = \bigcup_{i=1}^s \mathcal{V}_i \subseteq \mathbb{P}^n$ of dimension $n-k-1$ and degree $d$, where $\mathcal{V}_1, \ldots, \mathcal{V}_s$ are the irreducible components of $\mathcal{V}$, the Chow form of $\mathcal{V}$ is $R_{\mathcal{V}} := \prod_{i=1}^s R_{\mathcal{V}_i} \in (R/I)_d$. It is uniquely determined by $\mathcal{V}$ up to scalars.*

As a $k$-subspace intersects $\mathcal{V} = \bigcup_{i=1}^s \mathcal{V}_i$ iff it intersects some $\mathcal{V}_i$, we see from Theorem 3.7 that the Chow form $R_{\mathcal{V}}$ of an equidimensional projective subvariety $\mathcal{V}$ of dimension $n-k-1$ vanishes precisely at the set of $k$-subspaces that intersect $\mathcal{V}$.

# Original main statement (journal p.7)

For $n, d \in \mathbb{N}^+$ and $k \in \{0, 1, \ldots, n-1\}$, define $N(k,d,n)$ by

$$N(k,d,n) := \min\left\{ \binom{(k'+1)(n'+1+d)}{(k'+1)d}, \binom{(n-k)(n'+1+d)}{(n-k)d} \right\},$$

where $k' := \min\{k, d-2\} \leq k$ and $n' := k' + n - k \leq n$.

**Theorem 1.7 (Main Theorem).** *For $n, d \in \mathbb{N}^+$, $k \in \{0, 1, \ldots, n-1\}$, and $\epsilon \in (0,1)$, there exists an $(n,d,\epsilon)$-evasive $k$-subspace family (resp. affine $k$-subspace family) $\mathcal{H}$ on $\mathbb{P}^n$ (resp. $\mathbb{A}^n$) of size*

$$\mathrm{poly}(N(k,d,n), n, 1/\epsilon) \leq \mathrm{poly}(n^{\min\{k+1, n-k, d\}d}, d, 1/\epsilon).$$

*Moreover:*

- *If $\mathrm{char}(\mathbb{F}) = 0$, where $\mathbb{F}$ denotes the base field, then the linear equations defining the projective or affine subspaces in $\mathcal{H}$ are defined over $\mathbb{Q}$. Moreover, the bit-lengths of the numerators and denominators of the coefficients of these linear equations are polynomial in $|\mathcal{H}|$, and the total bit complexity of computing these linear equations is polynomial in $|\mathcal{H}|$.*
- *If $\mathrm{char}(\mathbb{F}) = p > 0$, then the linear equations defining the projective or affine subspaces in $\mathcal{H}$ are defined over an extension field $\mathbb{F}_q$ of $\mathbb{F}_p$, where $q \leq \mathrm{poly}(|\mathcal{H}|, p)$. The total bit complexity of computing these linear equations and constructing the field $\mathbb{F}_q$ is polynomial in $|\mathcal{H}|$ and $\log p$.*

*In particular, when $d$ is bounded, the bit complexity of constructing $\mathcal{H}$ is polynomial in $n/\epsilon$ (and $\log p$ if $\mathrm{char}(\mathbb{F}) = p > 0$).*

# Original degree-bounded evasion convention (journal p.6)

**Definition 1.5.** *We say a projective (resp. affine) $k$-subspace family $\mathcal{H}$ on $\mathbb{P}^n$ (resp. $\mathbb{A}^n$) is $(n, d)$-evasive if it is $\mathcal{F}$-evasive, where $\mathcal{F}$ is chosen to be the family of all subvarieties of $\mathbb{P}^n$ (resp. $\mathbb{A}^n$) of degree at most $d$. Similarly, we say $\mathcal{H}$ is $(n, d, \epsilon)$-evasive if it is $(\mathcal{F}, \epsilon)$-evasive.*

**Remark 1.6.** *In Definition 1.5, we do not make any assumption about the dimension of the varieties in $\mathcal{F}$ or their irreducible components. We will see in Section 3.1 that in fact, it suffices to consider the subfamily of equidimensional varieties or even irreducible varieties of dimension $n - k - 1$ when constructing variety evasive $k$-subspace families.*

# Original Section 2.1: Black-Box PIT for Low Degree Polynomials (journal pp.18–20)

**Definition 2.1 ($\epsilon$-hitting set).** *Let $\mathcal{F}$ be a family of polynomials in $\mathbb{F}[X_1, \ldots, X_n]$ and $\epsilon \in (0, 1)$. We say a finite collections of points $\mathcal{H} \subseteq \mathbb{F}^n$ is an $\epsilon$-hitting set for $\mathcal{F}$ if for any nonzero $Q \in \mathcal{F}$, the evaluation $Q(\alpha)$ is nonzero for all but at most $\epsilon$-fraction of $\alpha \in \mathcal{H}$.*

We need an explicit construction of $\epsilon$-hitting sets for low degree polynomials. This problem has been well studied (Bläser & Pandey 2020; Bogdanov 2005; Bshouty 2014; Cohen & Ta-Shma 2013; Demillo & Lipton 1978; Klivans & Spielman 2001; Lu 2012; Schwartz 1980; Zippel 1979). For completeness, we present a construction based on sparse polynomial identity testing. Recall that a polynomial is $s$-sparse if it has at most $s$ monomials. We need the following lemma from Agrawal et al. (2015).

**Lemma 2.2 (Agrawal et al. 2015, Lemma 4, restated).** *For $n, s, d \in \mathbb{N}^+$ and $\epsilon_0 \in (0, 1)$, there exist maps $w_1, w_2, \ldots, w_N : [n] \to [N \log N]$, where $N = \mathrm{poly}(n, s, \log d, \epsilon_0^{-1})$, such that for any nonzero $s$-sparse polynomial $f \in \mathbb{F}[X_1, \ldots, X_n]$ of individual degree at most $d$, all but at most $\epsilon_0$-fraction of $w_i$ among $w_1, \ldots, w_N$ satisfies $f(Y^{w_i(1)}, \ldots, Y^{w_i(n)}) \neq 0$. Moreover, the bit complexity of computing $w_1, w_2, \ldots, w_N$ is polynomial in $N$.*

Given $n, d \in \mathbb{N}^+$ and $\epsilon \in (0, 1)$, we construct an $\epsilon$-hitting set for $n$-variate polynomials of degree at most $d$ as follows:

1. Let $s = \binom{n+d}{d}$, $\epsilon_0 = \epsilon/2$, and $M = \lceil \epsilon_0^{-1} dN \log N \rceil$, where $N$ is as in Lemma 2.2.

2. Let $w_1, \ldots, w_N$ be as in Lemma 2.2, which can be computed in time $\mathrm{poly}(N)$.

3. If $\mathrm{char}(\mathbb{F}) = 0$, let $S = [M] \subseteq \mathbb{Z} \subseteq \mathbb{F}$. If $\mathrm{char}(\mathbb{F}) = p > 0$, choose the smallest $p$-power $q$ such that $q \geq M$, and choose $S$ to be a subset of $\mathbb{F}_q \subseteq \mathbb{F}$ of cardinality $M$. We remark that $\mathbb{F}_q$ can be constructed deterministically in time $\mathrm{poly}(M, \log p)$. To see this, note that $q \leq Mp$ by the minimality of $q$. If $M \leq p$, then $\mathbb{F}_q$ is just $\mathbb{F}_p$. On the other hand, if $p < M \leq q$, then $\mathbb{F}_q$ can be constructed in time $\mathrm{poly}(p, [\mathbb{F}_q : \mathbb{F}_p])$ (see, e.g., Lenstra (1990)), which is polynomial in $M$ since $p < M$ and $q \leq Mp$.

4. Finally, construct the following collection of points in $\mathbb{F}^n$ of size $MN$

$$T = \{(\alpha^{w_i(1)}, \ldots, \alpha^{w_i(n)}) : \alpha \in S, i \in [N]\} \subseteq \mathbb{F}^n.$$

**Lemma 2.3.** *For any nonzero polynomial $f \in \mathbb{F}[X_1, \ldots, X_n]$ of degree at most $d$, we have $f(u) \neq 0$ for all but at most $\epsilon$-fraction of $u \in T$. The collection $T$ has cardinality* $\mathrm{poly}(\binom{n+d}{d}, 1/\epsilon)$ *and can be computed in time* $\mathrm{poly}(|T|)$.

*Proof.* Let $f \in \mathbb{F}[X_1, \ldots, X_n]$ be a nonzero polynomial of degree at most $d$. Note that $f$ is trivially $s$-sparse, where $s = \binom{n+d}{d}$. So by Lemma 2.2, for all but at most $\epsilon_0$-fraction of $i \in [N]$, we have $\widetilde{f_i} := f(Y^{w_i(1)}, \ldots, Y^{w_i(n)}) \neq 0$. Consider $i \in [N]$ such that $\widetilde{f_i} \neq 0$. Note that $\widetilde{f_i}$ is a univariate polynomial of degree at most $dN \log N$. So it has at most $dN \log N \leq \epsilon_0 M$ zeros. Therefore, by the choice of $M$, we have $f(\alpha^{w_i(1)}, \ldots, \alpha^{w_i(n)}) = \widetilde{f_i}(\alpha) \neq 0$ for all but at most $\epsilon_0$-fraction of $\alpha \in S$. It follows that $f(u) \neq 0$ holds for all but at most $\epsilon$-fraction of $u \in T$, as claimed. The rest of the lemma follows easily from the construction. □

Note that the seed length required to choose a random element in $T$ is $\log |T| = O(\log \binom{n+d}{d} + \log(1/\epsilon))$, which is optimal up to a constant factor. We have made no effort to optimize the constant hidden in $O(\cdot)$. Interested readers may find the state-of-the-art result in Bläser & Pandey (2020), which achieves the optimal constant, at least when $d = n^{o(1)}$.

# Original Section 2.2: Explicit Lossless Rank Condensers (journal pp.20–21)

We need the following lemma in the context of lossless rank condensers. The construction in the lemma was given by Forbes & Shpilka (2012) and the lemma itself follows implicitly from the analysis of Forbes et al. (2014). It was also stated explicitly as Forbes (2014, Theorem 5.4.3).

**Lemma 2.4 (Forbes 2014; Forbes et al. 2014).** *Let $n \in \mathbb{N}^+$ and $r \in [n]$. Let $\omega \in \mathbb{F}^\times$ such that the multiplicative order of $\omega$ is at least $n$. Define the $r \times n$ matrix $W = (w_{i,j})_{i \in [r], j \in [n]}$ over $\mathbb{F}[X]$ by*

$$w_{i,j} = (\omega^{i-1} X)^{j-1}.$$

*Then for every $n \times r$ matrix $M$ over $\mathbb{F}$ of rank $r$, the polynomial $\det(WM) \in \mathbb{F}[X]$ is nonzero and has degree at most $r(n-r)$ after dividing out powers of $X$.*

**Corollary 2.5.** *Let $n, r, W$ be as in Lemma 2.4 and $\epsilon \in (0,1)$. Let $S \subseteq \mathbb{F}^\times$ be a finite set of cardinality at least $r(n-r)/\epsilon$. For every $n \times r$ matrix $M$ over $\mathbb{F}$ of rank $r$, we have* $\mathrm{rank}(W(\alpha)M) = r$ *for all but at most $\epsilon$-fraction of $\alpha \in S$, where $W(\alpha)$ denotes the matrix $(w_{i,j}(\alpha))_{i\in[r],j\in[n]}$ over $\mathbb{F}$.*

Corollary 2.5 states that the collection $\{W(\alpha) : \alpha \in S\}$ of matrices is a (weak) $(r, \epsilon|S|)$-lossless rank condenser, as defined in Forbes & Guruswami (2015). Note that for each $\alpha \in S$, we have $\mathrm{rank}(W(\alpha)) = r$ and hence $W(\alpha)$ corresponds to an $(r-1)$-subspace $U_{W(\alpha)}$ of $\mathbb{P}^{n-1}$. As explained in Introduction, the collection $\mathcal{H} = \{U_{W(\alpha)} : \alpha \in S\}$ is an $(\mathcal{F}, \epsilon)$-evasive $(r-1)$-subspace family on $\mathbb{P}^{n-1}$, where $\mathcal{F}$ is the family of $(n-r-1)$-subspaces of $\mathbb{P}^{n-1}$. Choosing $S$ of size $r(n-r)+1$ and $\epsilon = 1 - \frac{1}{r(n-r)+1}$ shows that the lower bound in Theorem 1.10 is achieved when $d = 1$.

# Original geometry background needed for the construction (journal pp.23–25)

**Morphisms.** Let $\mathcal{V}_1 \subseteq \mathbb{A}^n$ and $\mathcal{V}_2 \subseteq \mathbb{A}^m$ be affine varieties. A morphism from $\mathcal{V}_1$ to $\mathcal{V}_2$ is a map $f : \mathcal{V}_1 \to \mathcal{V}_2$ that is a restriction of a polynomial map $\mathbb{A}^n \to \mathbb{A}^m$. Such a morphism $f$ is associated with a ring homomorphism $f^\sharp : \mathbb{F}[\mathcal{V}_2] \to \mathbb{F}[\mathcal{V}_1]$, making $\mathbb{F}[\mathcal{V}_1]$ an algebra over $\mathbb{F}[\mathcal{V}_2]$. We say $f$ is finite if $\mathbb{F}[\mathcal{V}_1]$ is finitely generated as an $\mathbb{F}[\mathcal{V}_2]$-module. Let $f : \mathcal{V}_1 \to \mathcal{V}_2$ be a map between projective varieties $\mathcal{V}_1$ and $\mathcal{V}_2$. We say $f$ is a morphism from $\mathcal{V}_1$ to $\mathcal{V}_2$ if there exists a collection of open subsets $\{U_i\}_{i\in I}$ of $\mathcal{V}_2$ such that $\mathcal{V}_2 = \bigcup_{i\in I} U_i$ (i.e., $\{U_i\}_{i\in I}$ is an open cover of $\mathcal{V}_2$) and for each $i \in I$, the restriction $f|_{f^{-1}(U_i)} : f^{-1}(U_i) \to U_i$ is a morphism between affine varieties. Furthermore, if each $f|_{f^{-1}(U_i)}$ is finite, then we say $f$ is finite. Finiteness does not depend on the choice of the affine open cover. Namely, if $f : \mathcal{V}_1 \to \mathcal{V}_2$ is a finite morphism between projective varieties $\mathcal{V}_1$ and $\mathcal{V}_2$, and $U$ is an open subset of $\mathcal{V}_2$ such that $f|_{f^{-1}(U)} : f^{-1}(U) \to U$ is a morphism between affine varieties, then $f|_{f^{-1}(U)}$ is also finite. The image of a morphism $f : \mathcal{V}_1 \to \mathcal{V}_2$ is denoted by $\mathrm{Im}(f)$ or $f(\mathcal{V}_1)$. The image of a closed set under a finite morphism is still closed. The composition of two finite morphisms is still finite.

**Dimension.** The dimension of an irreducible variety $\mathcal{V}$, denoted by $\dim(\mathcal{V})$, is the largest integer $m$ such that there exists a chain of irreducible varieties $\emptyset \subsetneq \mathcal{V}_0 \subsetneq \mathcal{V}_1 \subsetneq \cdots \subsetneq \mathcal{V}_m = \mathcal{V}$. More generally, the dimension of a nonempty variety is the maximal dimension of its irreducible components. We define the dimension of an empty set to be $-\infty$. A variety is equidimensional if its irreducible components have the same dimension. If $\pi : \mathcal{V} \to \mathcal{V}'$ is a finite morphism, then $\dim(\mathcal{V}) = \dim(\pi(\mathcal{V}))$.

**Projective closure.** The affine $n$-space $\mathbb{A}^n$ may be regarded as an open subset of $\mathbb{P}^n$ via the map $(x_1, \ldots, x_n) \mapsto (x_1, \ldots, x_n, 1)$. The complement $H_\infty := \mathbb{P}^n \setminus \mathbb{A}^n$ is a hyperplane of $\mathbb{P}^n$ defined by $X_{n+1} = 0$, called the hyperplane at infinity. For an affine subvariety $\mathcal{V}$ of $\mathbb{A}^n \subseteq \mathbb{P}^n$, the smallest projective subvariety of $\mathbb{P}^n$ containing $\mathcal{V}$ is the projective closure of $\mathcal{V}$, which we denote by $\mathcal{V}^{\mathrm{cl}}$. It is known that $\mathcal{V}^{\mathrm{cl}} \cap \mathbb{A}^n = \mathcal{V}$, $\dim(\mathcal{V}^{\mathrm{cl}}) = \dim(\mathcal{V})$, and $\deg(\mathcal{V}^{\mathrm{cl}}) = \deg(\mathcal{V})$.

**Joins of disjoint projective varieties.** For two distinct points $p, q \in \mathbb{P}^n$, denote by $\overline{pq}$ the unique projective line passing through them. For two disjoint projective subvarieties $\mathcal{V}_1, \mathcal{V}_2 \subseteq \mathbb{P}^n$, define the join $J(\mathcal{V}_1, \mathcal{V}_2)$ of $\mathcal{V}_1$ and $\mathcal{V}_2$ as

$$J(\mathcal{V}_1, \mathcal{V}_2) := \bigcup_{p \in \mathcal{V}_1, q \in \mathcal{V}_2} \overline{pq}.$$

**Lemma 2.6 (Harris 1992, Examples 6.17, 11.36, and 18.17).** *$J(\mathcal{V}_1, \mathcal{V}_2)$ is a subvariety of $\mathbb{P}^n$ of dimension $\dim(\mathcal{V}_1) + \dim(\mathcal{V}_2) + 1$ and degree at most $\deg(\mathcal{V}_1) \cdot \deg(\mathcal{V}_2)$.*

**Lemma 2.7 (Harris 1992, Exercise 11.6 and Corollary 18.5).** *Let $\mathcal{V}$ be a nonempty equidimensional subvariety of $\mathbb{P}^n$ and $H$ a hypersurface of $\mathbb{P}^n$ not containing an irreducible component of $\mathcal{V}$. Then $\mathcal{V} \cap H$ is an equidimensional subvariety of dimension $\dim(\mathcal{V}) - 1$ and degree at most $\deg(\mathcal{V}) \cdot \deg(H)$ (or an empty set if $\dim(\mathcal{V}) = 0$).*

**Corollary 2.8.** *Let $\mathcal{V}$ be a subvariety of $\mathbb{P}^n$ of dimension $r$, where $0 \leq r < n$. Then there exists an $(n - r - 1)$-subspace $W$ disjoint from $\mathcal{V}$.*

*Proof.* It suffices to show that there exist hyperplanes $H_1, \ldots, H_{r+1}$, such that $\mathcal{V}_i := \mathcal{V} \cap (\bigcap_{j=1}^{i} H_j)$ is empty for some $i \leq r + 1$. We may inductively choose each $H_i$ such that $C \not\subseteq H_i$ for every irreducible component $C$ of $\mathcal{V}_{i-1}$, so that $\dim(\mathcal{V}_i) \leq \dim(\mathcal{V}_{i-1}) - 1$ by Lemma 2.7. So $\mathcal{V}_i = \emptyset$ for some $i \leq r + 1$. □

**Lemma 2.9 (Shafarevich 1994, Section I.6.2, Theorem 6).** *Suppose $\mathcal{V}_1$ and $\mathcal{V}_2$ are subvarieties of $\mathbb{P}^n$ and $\dim(\mathcal{V}_1) + \dim(\mathcal{V}_1) \geq n$. Then $\mathcal{V}_1 \cap \mathcal{V}_2 \neq \emptyset$ and $\dim(\mathcal{V}_1 \cap \mathcal{V}_2) \geq \dim(\mathcal{V}_1) + \dim(\mathcal{V}_1) - n$.*

# Original Section 3.1: Reducing to the Case of Equidimensional or Irreducible Varieties (journal pp.26–27)

The following lemma states that to construct $k$-subspace families that are evasive for subvarieties of $\mathbb{P}^n$, it suffices to consider equidimensional subvarieties of dimension $n - k - 1$ (i.e., codimension $k + 1$).

**Lemma 3.1.** *Let $n, d \in \mathbb{N}^+$ and $k \in \{0, 1, \ldots, n-1\}$. Let $\mathcal{F}$ be the family of all equidimensional subvarieties of $\mathbb{P}^n$ of dimension $n - k - 1$ and degree at most $d$. Then an $(\mathcal{F}, \epsilon)$-evasive $k$-subspace family is also $(n, d, \epsilon)$-evasive.*

The proof of Lemma 3.1 is based on the following claim.

**Claim 3.2.** *Let $\mathcal{V}$ be an irreducible subvariety of $\mathbb{P}^n$. There exists a subvariety $\widetilde{\mathcal{V}} \subseteq \mathbb{P}^n$ of dimension $n - k - 1$ and degree at most $\deg(\mathcal{V})$ such that any $k$-subspace of $\mathbb{P}^n$ that evades $\widetilde{\mathcal{V}}$ also evades $\mathcal{V}$.*

*Proof.* If $\dim(\mathcal{V}) = n-k-1$, then just let $\widetilde{\mathcal{V}} = \mathcal{V}$. Now assume $\dim(\mathcal{V}) < n-k-1$. Let $t = (n-k-1) - \dim(\mathcal{V}) - 1$ and let $\widetilde{\mathcal{V}}$ be the join of $\mathcal{V}$ and a $t$-subspace disjoint from $\mathcal{V}$ (which exists by Corollary 2.8). Then $\widetilde{\mathcal{V}}$ is a projective subvariety of dimension $n-k-1$ and degree at most $\deg(\mathcal{V})$ by Lemma 2.6. Suppose $W$ is a $k$-subspace that evades $\widetilde{\mathcal{V}}$. Then $W$ is disjoint from $\widetilde{\mathcal{V}} \supseteq \mathcal{V}$. So $W$ also evades $\mathcal{V}$. Finally, assume $\dim(\mathcal{V}) > n-k-1$. Let $t = \dim(\mathcal{V}) - (n-k-1)$. By Lemma 2.7, there exist $t$ hyperplanes $H_1, \ldots, H_t$ of $\mathbb{P}^n$ such that $\mathcal{V} \cap \bigcap_{i=1}^t H_i$ is equidimensional of dimension $n-k-1$ and degree at most $\deg(\mathcal{V})$. Let $\widetilde{\mathcal{V}} = \mathcal{V} \cap \bigcap_{i=1}^t H_i$. Suppose $W$ is a $k$-subspace that evades $\widetilde{\mathcal{V}}$. Then $W \cap \widetilde{\mathcal{V}} = (W \cap \mathcal{V}) \cap \bigcap_{i=1}^t H_i = \emptyset$. Again by Lemma 2.7, we have $\dim(W \cap \mathcal{V}) \leq t-1 = \dim(\mathcal{V}) + \dim(W) - n$. So $W$ also evades $\mathcal{V}$. □

*Proof of Lemma 3.1.* Consider a projective subvariety $\mathcal{V} \subseteq \mathbb{P}^n$ of degree at most $d$. Let $\mathcal{V}_1, \ldots, \mathcal{V}_s$ be the irreducible components of $\mathcal{V}$. For each $i \in [s]$, use Claim 3.2 to choose a projective subvariety $\widetilde{\mathcal{V}}_i \subseteq \mathbb{P}^n$ of dimension $n-k-1$ and degree at most $\deg(\mathcal{V}_i)$ such that any $k$-subspace that evades $\widetilde{\mathcal{V}}_i$ also evades $\mathcal{V}_i$. Let $\widetilde{\mathcal{V}} = \bigcup_{i=1}^s \widetilde{\mathcal{V}}_i$. Then $\widetilde{\mathcal{V}} \in \mathcal{F}$. By construction, any $k$-subspace that evades $\widetilde{\mathcal{V}}$ also evades $\mathcal{V}$. It follows that an $(\mathcal{F}, \epsilon)$-evasive $k$-subspace family is also $(n, d, \epsilon)$-evasive. □

We further reduce to the case of irreducible varieties at the cost of blowing up the parameter $\epsilon$ by a factor of $d$. This is useful as we need irreducibility later in Lemma 3.13.

**Lemma 3.3.** *Let $n, d \in \mathbb{N}^+$ and $k \in \{0, 1, \ldots, n-1\}$. Let $\mathcal{F}'$ be the family of all irreducible subvarieties of $\mathbb{P}^n$ of dimension $n-k-1$ and degree at most $d$. Then an $(\mathcal{F}', \epsilon)$-evasive $k$-subspace family is also an $(n, d, d\epsilon)$-evasive $k$-subspace family.*

*Proof.* Let $\mathcal{F}$ be as in Lemma 3.1. Each $\mathcal{V} \in \mathcal{F}$ has at most $d$ irreducible components, which are all in $\mathcal{F}'$ since their degrees are bounded by $d$. By definition and the union bound, if a $k$-subspace family $\mathcal{H}$ is $(\mathcal{F}', \epsilon)$-evasive, then it is also $(\mathcal{F}, d\epsilon)$-evasive. Combining this with Lemma 3.1 proves the lemma. □

# Original Chow forms in Stiefel coordinates (journal pp.31–32)

We may also express the Chow form in Stiefel coordinates, i.e., in the entries of a generating matrix of a linear subspace. This expression has the advantage that it is an actual polynomial rather than a member of the abstract vector space $(R/I)_d$. Formally, let $A^*$ be a $(k+1) \times (n+1)$ variable matrix whose $(i,j)$-th entry is a variable $Y_{i,j}$. Define the ring homomorphism

$$\phi^\sharp : R = \mathbb{F}[X_S : S \in \binom{[n+1]}{k+1}] \to \mathbb{F}[Y_{i,j} : i \in [k+1], j \in [n+1]]$$

that sends each variable $X_S$ to $\det(A^*_{[k+1],S})$. Define the Chow form of $\mathcal{V}$ in Stiefel coordinates to be

$$\widetilde{R}_{\mathcal{V}} := \phi^\sharp(P_{\mathcal{V}}) \in \mathbb{F}[Y_{i,j} : i \in [k+1], j \in [n+1]]$$

where $P_{\mathcal{V}} \in R_d$ is a lift of $R_{\mathcal{V}} \in (R/I)_d$. Note that $I$ is precisely the kernel of $\phi^\sharp$. So $\widetilde{R}_{\mathcal{V}}$ is uniquely determined by $\mathcal{V}$ up to scalars. By construction, for any $W \in \mathrm{G}(k+1, n+1)$ and generating matrix $A = (a_{i,j})_{i\in[k+1],j\in[n+1]}$ of $W$, we have $P_{\mathcal{V}}(\phi(W)) = \widetilde{R}_{\mathcal{V}}(A) := \widetilde{R}_{\mathcal{V}}(a_{1,1}, \ldots, a_{k+1,n+1})$. So $\widetilde{R}_{\mathcal{V}}$ vanishes at $A$ iff $\mathbb{P}W \in \mathrm{G}(k, n)$ intersects $\mathcal{V}$.

**Chow forms in dual Stiefel coordinates.** Similarly, we may express the Chow form in dual Stiefel coordinates, i.e., in the entries of a parity check matrix of a linear subspace. More specifically, choose a homogeneous polynomial $Q_{\mathcal{V}} \in \mathbb{F}[X_S : S \in \binom{[n+1]}{n-k}]$ that defines the set of $k$-subspaces intersecting $\mathcal{V}$ in terms of dual Plücker coordinates. As primal and dual Plücker coordinates are equivalent, $Q_{\mathcal{V}}$ can be obtained from the polynomial $P_{\mathcal{V}}$ above by simply negating and renaming variables. Next, compose $Q_{\mathcal{V}}$ with a ring homomorphism that substitutes dual Plücker coordinates with dual Stiefel coordinates. The resulting polynomial, which we denote by $\widetilde{R}^{\vee}_{\mathcal{V}} \in \mathbb{F}[Y_{i,j} : i \in [n-k], j \in [n+1]]$, is called the Chow form of $\mathcal{V}$ in dual Stiefel coordinates. We note that the Chow form $\widetilde{R}_{\mathcal{V}}$ in primal Stiefel coordinates is a homogeneous polynomial of degree $(k+1)d$ in $(k+1)(n+1)$ variables, whereas the Chow form $\widetilde{R}^{\vee}_{\mathcal{V}}$ in dual Stiefel coordinates is a homogeneous polynomial of degree $(n-k)d$ in $(n-k)(n+1)$ variables. This suggests that it is more convenient to use the Chow form in primal (resp. dual) Stiefel coordinates when $k$ is small (resp. $n-k$ is small).

# Original Section 3.3: Explicit Constructions of Variety Evasive Subspace Families (journal pp.32–40)

Let $n, d \in \mathbb{N}^+$, $k \in \{0, 1, \ldots, n-1\}$, and $\epsilon \in (0, 1)$. In this subsection, we prove the Main theorem (Theorem 1.7) by constructing explicit projective or affine $k$-subspace families that are $(n, d, \epsilon)$-evasive. We first prove Theorem 1.7 in the projective case and then derive the affine case from it by viewing $\mathbb{A}^n$ as an open subset of $\mathbb{P}^n$. For the projective case, we present two constructions. The first one is simple and only uses $\epsilon$-hitting sets for low degree polynomials (Lemma 2.3). But the size of the resulting subspace family is polynomial only when both $d$ and $k$ (or $n-k$) are bounded. Next, we give a more sophisticated construction, which yields subspace families of polynomial size as long as $d$ is bounded.

## 3.3.1 Simple Construction

We first present a simple construction of $(n, d, \epsilon)$-evasive $k$-subspace families on $\mathbb{P}^n$. First assume $k+1 \le n-k$. In this case, construct a $k$-subspace family $\mathcal{H}$ on $\mathbb{P}^n$ as follows:

1. Use Lemma 2.3 to compute an $\epsilon$-hitting set $T$ for the family of polynomials $f \in \mathbb{F}[Y_{i,j} : i \in [k+1], j \in [n+1]]$ of degree at most $(k+1)d$ such that $|T| = \mathrm{poly}(\binom{(k+1)(n+1+d)}{(k+1)d}, 1/\epsilon)$. Think of $T$ as a collection of $(k+1) \times (n+1)$ matrices over $\mathbb{F}$.

2. Initialize $\mathcal{H} = \emptyset$. For each matrix $A \in T$, if $A$ has full row rank $k+1$, add to $\mathcal{H}$ the $k$-subspace $W \in \mathrm{G}(k,n)$ with the generating matrix $A$.

Next, assume $k+1 > n-k$. In this case, construct $\mathcal{H}$ in a similar way, but use parity check matrices instead of generating matrices. Namely, compute an $\epsilon$-hitting set $T$ for the family of polynomials $f \in \mathbb{F}[Y_{i,j} : i \in [n-k], j \in [n+1]]$ of degree at most $(n-k)d$ such that $|T| = \mathrm{poly}(\binom{(n-k)(n+1+d)}{(n-k)d}, 1/\epsilon)$. Think of $T$ as a collection of $(n-k) \times (n+1)$ matrices over $\mathbb{F}$. For each matrix $A \in T$, add to $\mathcal{H}$ the $k$-subspace $W \in \mathrm{G}(k,n)$ with the parity check matrix $A$. This construction does give an $(n,d,\epsilon)$-evasive $k$-subspace family, as stated by the following lemma.

**Lemma 3.11.** *The $k$-subspace family $\mathcal{H}$ constructed above is $(n,d,\epsilon)$-evasive and has size polynomial in*

$$\min\left\{\binom{(k+1)(n+1+d)}{(k+1)d}, \binom{(n-k)(n+1+d)}{(n-k)d}\right\}$$

*and $1/\epsilon$. Moreover, the total time complexity of computing the linear equations defining the $k$-subspaces in $\mathcal{H}$ is polynomial in $|\mathcal{H}|$ (and $\log p$, if $\mathrm{char}(\mathbb{F}) = p > 0$).*

*Proof.* We only show that $\mathcal{H}$ is $(n,d,\epsilon)$-evasive since the rest of the lemma is obvious from the construction. Let $\mathcal{F}$ be the family of all equidimensional subvarieties of $\mathbb{P}^n$ of dimension $n-k-1$ and degree at most $d$. By Lemma 3.1, it suffices to prove that $\mathcal{H}$ is $(\mathcal{F},\epsilon)$-evasive. Consider any $\mathcal{V} \in \mathcal{F}$. We want to show that $\mathcal{V} \cap W = \emptyset$ for all but at most $\epsilon$-fraction of $W \in \mathcal{H}$. First assume $k+1 \le n-k$, or equivalently, $\binom{(k+1)(n+1+d)}{(k+1)d} \le \binom{(n-k)(n+1+d)}{(n-k)d}$. The Chow form $\widetilde{R}_{\mathcal{V}}$ of $\mathcal{V}$ in Stiefel coordinates is a nonzero homogeneous polynomial in $\mathbb{F}[Y_{i,j} : i \in [k+1], j \in [n+1]]$ of degree $(k+1)\deg(\mathcal{V}) \le (k+1)d$. By the choice of $T$, for all but at most $\epsilon$-fraction of $A \in T$, we have $\widetilde{R}_{\mathcal{V}}(A) \neq 0$, which implies $\mathcal{V} \cap W = \emptyset$, where $A$ is a generating matrix of $W$. By construction, $\mathcal{H}$ is the collection of $k$-subspaces corresponding to the matrices $A \in T$ of full row rank. So we have ignored the matrices that do not have full row rank. But this does not increase the fraction of "bad" $W \in \mathcal{H}$ since if $A$ does not have full row rank, then the maximal minors of $A$ are all zero, and $\widetilde{R}_{\mathcal{V}}(A)$ must be zero. It follows that $\mathcal{V} \cap W = \emptyset$ for all but at most $\epsilon$-fraction of $W \in \mathcal{H}$, as desired. Now assume $k+1 > n-k$. The proof in this case is similar and we omit the details. The only difference is that we use the Chow form $\widetilde{R}^{\vee}_{\mathcal{V}}$ in dual Stiefel coordinates instead of $\widetilde{R}_{\mathcal{V}}$. □

### 3.3.2 Improved Construction

Before presenting the improved construction, we first introduce some notions from algebraic geometry.

**Projections.** Suppose $W$ is a $k$-subspace of $\mathbb{P}^n$, and $\ell_1, \ldots, \ell_{n-k} \in \mathbb{F}[X_1, \ldots, X_{n+1}]$ are $n-k$ homogeneous linear polynomials such that $W = V(\ell_1, \ldots, \ell_{n-k})$. Then we have a map $\pi : \mathbb{P}^n \setminus W \to \mathbb{P}^{n-k-1}$ defined by

$$\pi : x \mapsto (\ell_1(x), \ldots, \ell_{n-k}(x))$$

which is well-defined since $\ell_1, \ldots, \ell_{n-k}$ never simultaneously vanish on $\mathbb{P}^n \setminus W$. We say $\pi$ is a projection from $\mathbb{P}^n \setminus W$ to $\mathbb{P}^{n-k-1}$ and $W$ is its center. Note that if we lift $\pi$ to the linear map $\pi' : \mathbb{A}^{n+1} \to \mathbb{A}^{n-k}$ sending $x \in \mathbb{A}^{n+1}$ to $(\ell_1(x), \ldots, \ell_{n-k}(x)) \in \mathbb{A}^{n-k}$, then the center $W$ is simply $\mathbb{P}\ker(\pi')$. We need the following lemma.

**Lemma 3.12 (Shafarevich 1994, Section I.5.3, Theorem 7).** *Suppose $\pi : \mathbb{P}^n \setminus W \to \mathbb{P}^m$ is a projection with center $W$ and $\mathcal{V}$ is a subvariety of $\mathbb{P}^n$ disjoint from $W$. Then $\pi$ restricts to a finite morphism from $\mathcal{V}$ to $\mathbb{P}^m$.*

**Nondegenerate varieties.** For a subvariety $\mathcal{V} \subseteq \mathbb{P}^n$, denote by $\mathrm{span}(\mathcal{V})$ the smallest projective subspace that contains $\mathcal{V}$. We say $\mathcal{V}$ is nondegenerate if it is not contained in a hyperplane of $\mathbb{P}^n$, or equivalently, $\mathrm{span}(\mathcal{V}) = \mathbb{P}^n$. We need the following fact from algebraic geometry (see, e.g., Eisenbud & Harris (1987, Proposition 0) or Harris (1992, Corollary 18.12)).

**Lemma 3.13.** *The codimension of a nondegenerate irreducible subvariety $\mathcal{V}$ of $\mathbb{P}^n$ is at most* $\deg(\mathcal{V}) - 1$.

**A two-step construction.** We now give an improved construction of $(n, d, \epsilon)$-evasive $k$-subspace families on $\mathbb{P}^n$ as follows.

1. If $k \le d-2$, just use the previous simple construction. So assume $k > d-2$. Let $k' = d - 2 < k$, $n' = k' + n - k < n$, and $\epsilon_0 = \epsilon/(2d)$.
2. Use Corollary 2.5 to construct a collection $\mathcal{H}_1$ of $(n'+1) \times (n+1)$ matrix over $\mathbb{F}$ such that $|\mathcal{H}_1| = \mathrm{poly}(n, d/\epsilon)$ and for every $(n+1) \times (n'+1)$ matrix $M$ over $\mathbb{F}$ of rank $n'+1$, all but at most $\epsilon_0$-fraction of $B \in U$ satisfies $\mathrm{rank}(BM) = n'+1$. We abuse the notation and view $\mathcal{H}_1$ as a collection of linear maps from $\mathbb{A}^{n+1}$ to $\mathbb{A}^{n'+1}$. Then for any linear subspace $W \subseteq \mathbb{A}^{n+1}$ of dimension $n'+1$, we have $\dim(\pi(W)) = \dim(W) = n'+1$ for all but at most $\epsilon_0$-fraction of $\pi \in \mathcal{H}_1$.
3. Construct a collection $\mathcal{H}_2$ of linear maps from $\mathbb{A}^{n'+1}$ to $\mathbb{A}^{n-k}$ as follows. First assume $d > 1$. Use Lemma 3.11 to construct an $(n', d, \epsilon_0)$-evasive $k'$-subspace family $\mathcal{H}_2'$ on $\mathbb{P}^{n'}$ of size polynomial in
$$\min\left\{\binom{(k'+1)(n'+1+d)}{(k'+1)d}, \binom{(n-k)(n'+1+d)}{(n-k)d}\right\}$$
and $1/\epsilon_0$. For each $k'$-subspace $W \in \mathcal{H}_2$, compute a surjective linear map $\pi_W : \mathbb{A}^{n'+1} = \mathbb{A}^{n-k+k'+1} \to \mathbb{A}^{n-k}$ such that $W = \mathbb{P}\ker(\pi_W)$. Let $\mathcal{H}_2 = \{\pi_W : W \in \mathcal{H}_2'\}$. If $d = 1$, just let $\mathcal{H}_2$ be the singleton consisting of the identity map on $\mathbb{A}^{n'+1} = \mathbb{A}^{n-k}$.
4. Initialize $\mathcal{H} = \emptyset$. For each $(\pi_1, \pi_2) \in \mathcal{H}_1 \times \mathcal{H}_2$, if $\dim(\ker(\pi_2 \circ \pi_1)) = k+1$, add the $k$-subspace $\mathbb{P}\ker(\pi_2 \circ \pi_1)$ to $\mathcal{H}$.

We use the construction above to prove the Main theorem (Theorem 1.7) in the projective case. For convenience, we restate it in the following form.

**Theorem 3.14 (Main theorem in the projective case).** *The $k$-subspace family $\mathcal{H}$ constructed above is $(n, d, \epsilon)$-evasive and has size* $\mathrm{poly}(N(k, d, n), n, 1/\epsilon)$.

*Moreover, the total time complexity of computing the linear equations defining the $k$-subspaces in $\mathcal{H}$ is polynomial in $|\mathcal{H}|$ (and $\log p$, if $\mathrm{char}(\mathbb{F}) = p > 0$).*

*Proof.* The theorem follows from Lemma 3.11 if $k \leq d-2$. So assume $k > d-2$. We only show that $\mathcal{H}$ is $(n, d, \epsilon)$-evasive since the rest of the theorem is obvious from the construction. Let $\mathcal{F}$ be the family of all irreducible subvarieties of $\mathbb{P}^n$ of dimension $n-k-1$ and degree at most $d$. By Lemma 3.3, it suffices to prove that $\mathcal{H}$ is $(\mathcal{F}, 2\epsilon_0)$-evasive. Consider any $\mathcal{V} \in \mathcal{F}$. We want to show that $\mathcal{V} \cap W = \emptyset$ for all but at most $(2\epsilon_0)$-fraction of $W \in \mathcal{H}$. By definition, $\mathcal{V}$ is a nondegenerate irreducible subvariety of $\mathrm{span}(\mathcal{V})$. By Lemma 3.13, the codimension of $\mathcal{V}$ in $\mathrm{span}(\mathcal{V})$ is at most $d-1$. Therefore,

$$\dim(\mathrm{span}(\mathcal{V})) \leq \dim(\mathcal{V}) + d - 1 = (n-k-1) + (d-1) = n'.$$

Let $\Lambda \subseteq \mathbb{P}^n$ be an $n'$-subspace that contains $\mathrm{span}(\mathcal{V})$. By the choice of $\mathcal{H}_1$, all but at most $\epsilon_0$-fraction of $\pi_1 \in \mathcal{H}_1$ satisfies $\mathbb{P}\ker(\pi_1) \cap \Lambda = \emptyset$. Fix $\pi_1$ that satisfies this condition. Then $\mathbb{P}\ker(\pi_1)$ is disjoint from $\mathcal{V} \subseteq \Lambda$. By Lemma 3.12, $\pi_1$ induces a finite morphism $\overline{\pi}_1 : \mathcal{V} \to \mathbb{P}^{n'}$. Let $\mathcal{V}' = \overline{\pi}_1(\mathcal{V}) \subseteq \mathbb{P}^{n'}$. Then $\mathcal{V}'$ is a projective subvariety of dimension $\dim(\mathcal{V}) = n-k-1$ and degree at most $d$. (The degree bound follows from, e.g., an inductive application of Mumford (1976, Proposition 5.5).) By the choice of $\mathcal{H}_2$, all but at most $\epsilon_0$-fraction of $\pi_2 \in \mathcal{H}_2$ satisfies $\mathbb{P}\ker(\pi_2) \cap \mathcal{V}' = \emptyset$. Fix $\pi_2$ that satisfies this condition. (If $d = 1$ and $\pi_2$ is the identity map, we regard $\mathbb{P}\ker(\pi_2)$ as an empty set, in which case this condition is also satisfied.) By Lemma 3.12, $\pi_2$ induces a finite morphism $\overline{\pi}_2 : \mathcal{V}' \to \mathbb{P}^{n-k-1}$. So we have a finite morphism $\overline{\pi}_2 \circ \overline{\pi}_1 : \mathcal{V} \to \mathbb{P}^{n-k-1}$. Note that $\overline{\pi}_2 \circ \overline{\pi}_1$ is defined by restricting a projection with center $\mathbb{P}\ker(\pi_2 \circ \pi_1)$ to $\mathcal{V}$. As $\overline{\pi}_2 \circ \overline{\pi}_1$ is well-defined on $\mathcal{V}$, its center $\mathbb{P}\ker(\pi_2 \circ \pi_1)$ is disjoint from $\mathcal{V}$ and this also forces $\dim \ker(\pi_2 \circ \pi_1) = k+1$ by Lemma 2.9. By the above argument and the construction of $\mathcal{H}$, all but at most $2\epsilon_0$-fraction of the $k$-subspaces in $\mathcal{H}$ are disjoint from $\mathcal{V}$, as desired. □

# Original Section 3.3.4: Proof of the Main theorem in the Affine Case (journal pp.39–40)

We now prove Theorem 1.7 in the affine case. Recall that we may view $\mathbb{A}^n$ as an open subset of $\mathbb{P}^n$ via the map $(x_1, \ldots, x_n) \mapsto (x_1, \ldots, x_n, 1)$. In this way, $\mathbb{P}^n$ becomes the disjoint union of $\mathbb{A}^n$ and the hyperplane at infinity $H_\infty$ defined by $X_{n+1} = 0$. We use the following lemma to reduce the affine case to the projective case.

**Lemma 3.15.** *Let $n, d \in \mathbb{N}^+$, $k \in \{0, 1, \ldots, n-1\}$, and $\epsilon' \in (0, 1/2)$. Suppose $\mathcal{H}$ is an $(n, d, \epsilon')$-evasive $k$-subspace family on $\mathbb{P}^n$. Then*

$$\mathcal{H}' = \{W \cap \mathbb{A}^n : W \in \mathcal{H}, W \not\subseteq H_\infty\}$$

*is an $(n, d, \epsilon)$-evasive affine $k$-subspace family on $\mathbb{A}^n$, where $\epsilon = \epsilon'/(1-\epsilon') \leq 2\epsilon'$. Moreover,*

$$\mathcal{H}'' = \{W \in \mathcal{H} : W \not\subseteq H_\infty\} = \{W^{\mathrm{cl}} : W \in \mathcal{H}'\}$$

*is an $(n, d, \epsilon)$-evasive $k$-subspace family on $\mathbb{P}^n$.*

*Proof.* By $(n, d, \epsilon')$-evasiveness of $\mathcal{H}$, at most $\epsilon'$-fraction of $W \in \mathcal{H}$ are fully contained in $H_\infty$. Throwing away those $k$-subspaces fully contained in $H_\infty$ increases the error parameter $\epsilon'$ by at most a factor of $1/(1-\epsilon')$. Therefore, $\mathcal{H}'' = \{W \in \mathcal{H} : W \not\subseteq H_\infty\}$ is $(n, d, \epsilon)$-evasive. We want to prove that $\mathcal{H}' = \{W \cap \mathbb{A}^n : W \in \mathcal{H}''\}$ is also $(n, d, \epsilon)$-evasive. Consider a subvariety $\mathcal{V} \subseteq \mathbb{A}^n$ of degree at most $d$. Let $\mathcal{V}_1, \ldots, \mathcal{V}_s$ be the irreducible components of $\mathcal{V}$. The projective closure $\mathcal{V}^{\mathrm{cl}}$ of $\mathcal{V}$ has the irreducible components $(\mathcal{V}_1)^{\mathrm{cl}}, \ldots, (\mathcal{V}_s)^{\mathrm{cl}}$. Consider a $k$-subspace $W \in \mathcal{H}''$ that evades $\mathcal{V}^{\mathrm{cl}}$. We just need to prove that $W \cap \mathbb{A}^n$ evades $\mathcal{V}$. This is true since for each $i \in [s]$,

$$\begin{aligned}\dim((W \cap \mathbb{A}^n) \cap \mathcal{V}_i) &\le \dim(W \cap (\mathcal{V}_i)^{\mathrm{cl}}) \\ &\le \dim(W) + \dim((\mathcal{V}_i)^{\mathrm{cl}}) - n \\ &= \dim(W \cap \mathbb{A}^n) + \dim(\mathcal{V}_i) - n\end{aligned}$$

where the second inequality holds since $W$ evades $\mathcal{V}^{\mathrm{cl}}$ and the last equality uses the fact $W \not\subseteq H_\infty$. □

The affine case of Theorem 1.7 now follows easily.

*Proof of Theorem 1.7 in the affine case.* If $k = n$, just choose $\mathcal{H} = \mathbb{A}^n$. Now assume $k < n$. Construct an $(n, d, \epsilon/2)$-evasive $k$-subspace family $\mathcal{H}$ on $\mathbb{P}^n$ using Theorem 3.14. Then

$$\mathcal{H}' := \{W \cap \mathbb{A}^n : W \in \mathcal{H}, W \not\subseteq H_\infty\}$$

is an $(n, d, \epsilon)$-evasive affine $k$-subspace family on $\mathbb{A}^n$ by Lemma 3.15. The nonhomogeneous linear equations defining $W \cap \mathbb{A}^n \in \mathcal{H}'$ can be easily computed from the homogeneous linear equations defining $W \in \mathcal{H}$ by letting $X_{n+1} = 1$. □

The proof of Theorem 1.7 is now complete.

# Original Section 3.3.3: Derandomizing Noether's Normalization Lemma (journal pp.37–39)

We now prove Theorem 1.12 and Theorem 1.13. For convenience, we restate the theorems below.

**Theorem 1.12.** *Let $n, d \in \mathbb{N}^+$, $r \in \{0, 1, \ldots, n\}$, $k = n - r - 1$, and $\epsilon \in (0, 1)$. There exists an explicit collection $\mathcal{L}$ of linear maps $\mathbb{A}^{n+1} \to \mathbb{A}^{r+1}$ of size $\mathrm{poly}(N(k, d, n), n, 1/\epsilon)$ such that for every subvariety $\mathcal{V} \subseteq \mathbb{P}^n$ of dimension $r$ and degree at most $d$, all but at most $\epsilon$-fraction of $\pi \in \mathcal{L}$ induce a surjective finite morphism from $\mathcal{V}$ to $\mathbb{P}^r$.[1] Moreover, $\mathcal{L}$ can be computed in time polynomial in $|\mathcal{L}|$ (and $\log p$, if $\mathrm{char}(\mathbb{F}) = p > 0$).*

*Proof.* If $r = n$, just use the identity map on $\mathbb{A}^{n+1} = \mathbb{A}^{r+1}$. So assume $r < n$. Use Theorem 3.14 to construct an $(n, d, \epsilon)$-evasive $k$-subspace family $\mathcal{H}$ on $\mathbb{P}^n$ of size $\mathrm{poly}(N(k, d, n), n, 1/\epsilon)$. For each $k$-subspace $W \in \mathcal{H}$, compute a surjective linear map $\pi_W : \mathbb{A}^{n+1} \to \mathbb{A}^{r+1}$ such that $W = \mathbb{P}\ker(\pi_W)$. Let $\mathcal{L} = \{\pi_W : W \in \mathcal{H}\}$. Then $\mathcal{L}$ is a desired collection of linear maps by Lemma 3.12. □

[1] Let $N(k, d, n) = 1$ when $r = n$ (i.e., $k = -1$). Similarly, in Theorem 1.13, let $N(k, d, n-1) = 1$ when $r = n$ or $r = 0$ (i.e., $k = -1$ or $k = n - 1$).

**Theorem 1.13.** *Let $n, d \in \mathbb{N}^+$, $r \in \{0, 1, \ldots, n\}$, $k = n - r - 1$, and $\epsilon \in (0, 1)$. There exists an explicit collection $\mathcal{L}$ of linear maps $\mathbb{A}^n \to \mathbb{A}^r$ of size $\mathrm{poly}(N(k, d, n-1), n, 1/\epsilon)$ such that for every subvariety $\mathcal{V} \subseteq \mathbb{A}^n$ of dimension $r$ and degree at most $d$, all but at most $\epsilon$-fraction of $\pi \in \mathcal{L}$ restrict to a surjective finite morphism from $\mathcal{V}$ to $\mathbb{A}^r$. Moreover, $\mathcal{L}$ can be computed in time polynomial in $|\mathcal{L}|$ (and $\log p$, if $\mathrm{char}(\mathbb{F}) = p > 0$).*

*Proof.* If $r = n$, just use the identity map on $\mathbb{A}^n = \mathbb{A}^r$. If $r = 0$, use the only map $\mathbb{A}^n \to \mathbb{A}^0$. So assume $0 < r < n$. Regard $\mathbb{A}^n$ as an open subset of $\mathbb{P}^n$ via $(x_1, \ldots, x_n) \mapsto (x_1, \ldots, x_n, 1)$. Similarly, regard $\mathbb{A}^r$ as an open subset of $\mathbb{P}^r$ via $(x_1, \ldots, x_r) \mapsto (x_1, \ldots, x_r, 1)$. Let $H_\infty$ be the hyperplane at infinity of $\mathbb{P}^n$ defined by $X_{n+1} = 0$. Use Theorem 3.14 to construct an $(n-1, d, \epsilon)$-evasive $k$-subspace family $\mathcal{H}$ on $H_\infty \cong \mathbb{P}^{n-1}$ of size $\mathrm{poly}(N(k, d, n-1), n, 1/\epsilon)$. For each $W \in \mathcal{H}$, choose $n - k = r + 1$ homogeneous linear polynomials $\ell_1, \ldots, \ell_{r+1} \in \mathbb{F}[X_1, \ldots, X_{n+1}]$ such that $\ell_{r+1} = X_{n+1}$, $\ell_1, \ldots, \ell_r \in \mathbb{F}[X_1, \ldots, X_n]$, and $W = V(\ell_1, \ldots, \ell_{r+1})$. This is possible as $W \subseteq H_\infty = V(X_{n+1})$. These $r + 1$ linear polynomials determine a projection $\pi_W : \mathbb{P}^n \setminus W \to \mathbb{P}^r$, defined by

$$\begin{aligned} x = (x_1, \ldots, x_{n+1}) &\mapsto (\ell_1(x), \ldots, \ell_{r+1}(x)) \\ &= (\ell_1(x), \ldots, \ell_r(x), x_{n+1}). \end{aligned}$$

As $x_{n+1} = 1$ for $x \in \mathbb{A}^n$, we have $\pi_W(\mathbb{A}^n) \subseteq \mathbb{A}^r$. Restricting $\pi_W$ on $\mathbb{A}^n$ yields a map $\pi_W|_{\mathbb{A}^n} : \mathbb{A}^n \to \mathbb{A}^r$, which is a linear map as $\ell_1, \ldots, \ell_r$ are homogeneous linear polynomials in $\mathbb{F}[X_1, \ldots, X_n]$. Let $\mathcal{L} = \{\pi_W|_{\mathbb{A}^n} : W \in \mathcal{H}\}$. Let $\mathcal{V}$ be a subvariety of $\mathbb{A}^n$ of dimension $r$ and degree at most $d$. Its projective closure $\mathcal{V}^{\mathrm{cl}}$ has dimension $\dim(\mathcal{V}) = r$ and degree $\deg(\mathcal{V}) \leq d$. By the definition of $\mathcal{V}^{\mathrm{cl}}$, none of the irreducible components of $\mathcal{V}^{\mathrm{cl}}$ is fully contained in $H_\infty$. So by Lemma 2.7, the projective subvariety $\mathcal{V}^{\mathrm{cl}} \cap H_\infty$ has dimension $r - 1$ and degree at most $d$. By the choice of $\mathcal{H}$, all but at most $\epsilon$-fraction of $W \in \mathcal{H}$ are disjoint from $\mathcal{V}^{\mathrm{cl}} \cap H_\infty$ and hence from $\mathcal{V}^{\mathrm{cl}}$. So we just need to prove that for every $W \in \mathcal{H}$ disjoint from $\mathcal{V}^{\mathrm{cl}}$ and the corresponding projection $\pi_W$, the map $\pi_W|_{\mathcal{V}} : \mathcal{V} \to \mathbb{A}^r$ is a surjective finite morphism. This follows from Lemma 3.12 and the fact that $\mathcal{V} = \mathcal{V}^{\mathrm{cl}} \cap (\pi_W)^{-1}(\mathbb{A}^r)$. □

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